

Recall from previously:

(a) **Definition of inner product space**

1. A function on $V \times V$: $\langle u, v \rangle$
2. This is symmetric, linear in both u and v .
3. $\langle u, u \rangle > 0$ if $u \neq 0$.

(b) **Examples**

1. $\mathbb{R}^n$ and its subspaces with $\langle, \rangle = \text{dot product}$.
2. Polynomials of degree n and evaluation at $n + 1$ or more points:

$$\langle P, Q \rangle = P(t_0)Q(t_0) + \cdots + P(t_n)Q(t_n).$$

With regards to $\langle P, P \rangle > 0$, a polynomial that vanishes at $n + 1$ points is identically zero since it has at most n roots.

3. Continuous functions on $[a, b]$ with inner product $\langle f, g \rangle = \int_a^b f(t)g(t) dt$.

(c) **Length, distance, and orthogonality**

1. $\text{dist}(u, v) = \|u - v\|$.
2. Length of u is $\|u\|$.
3. v is said to be orthogonal to u if $\langle v, u \rangle = 0$.
4. Orthogonal complements of subspaces.
5. Orthogonal projection as before.

(d) **Gram-Schmidt.** This is the same as before for finite or countably infinite bases.

(e) **Best approximation.** If $W \subset V$ is a subspace with an orthogonal basis:

1. Orthogonal projection of u onto a subspace $W \subset V$ with an orthogonal basis is as before:

$$\text{proj}_W(u) = c_1v_1 + \cdots + c_kv_k, \quad \text{where} \quad c_k = \frac{\langle v_k, u \rangle}{\|v_k\|^2}.$$

2. $\text{proj}_W(u)$ is the closest vector in W to u (it minimizes the function $v \mapsto \|u - v\|^2$ for $v \in W$).

(f) **Inequalities**

1. Cauchy-Schwarz inequality: $|\langle u, v \rangle| \leq \|u\|\|v\|$.
2. Triangle inequality: $\|u - v\| \leq \|u\| + \|v\|$.
3. Angle between u and v : $\langle u, v \rangle = \|u\|\|v\|\cos(\theta)$.

Definition 1. An $n \times n$ matrix A is said to be symmetric when $A^T = A$.

1. True or false: A diagonal matrix is symmetric.

2. Prove the following theorem using the outline below.

Transpose Theorem: If u and v are vectors in $\mathbb{R}^n$ and A is an $n \times n$ matrix, then $u \cdot (Av) = (A^T u) \cdot v$.

a) Take a basis E_{ij} for $n \times n$ matrices with E_{ij} having all entries zero except for the i, j entry which is 1. Verify the theorem for E_{ij} .

b) Since the equation $u \cdot (Av) - (A^T u) \cdot v = 0$ is a linear equation for the components of A for any fixed choice of u and v , it is sufficient to verify the equation for each basis element E_{ij} . (Justify this statement!)

3. True or false: An orthogonally diagonalizable matrix is symmetric, where orthogonally diagonalizable is defined as follows:

Definition 2. An $n \times n$ matrix A is said to be orthogonally diagonalizable if there is a diagonal matrix D and orthogonal matrix P with $P^T = P^{-1}$ such that $A = PDP^{-1}$.

Theorem: A matrix is orthogonally diagonalizable if and only if it is symmetric.

- a) Thus, A has n real eigenvalues counting multiplicity.
- b) The eigenvectors form a complete, orthogonal basis.

4. Find the eigenvalues and then an orthonormal basis of eigenvectors for each of the matrices below:

a) $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$

5. True or false: If A is any given $m \times n$ matrix, then $A^T A$ is a symmetric, $n \times n$ matrix.

6. True or false: If A is any given $m \times n$ matrix, then the eigenvalues of $A^T A$ are the same as those of AA^T .

7. Let A denote the 3×1 matrix $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

a) Find the eigenvalues of $A^T A$.

b) Find the eigenvalues of AA^T .

8. Let u_1 and u_2 denote vectors in $\mathbb{R}^3$. Let $A = u_1 u_1^T + u_2 u_2^T$.

a) True or false: The matrix A always has 0 for one of its eigenvalues.

b) True or false: The eigenvalue 0 has multiplicity 1 if and only if u_1 and u_2 are not linearly independent.

9. **Spectral theorem** Let A be a symmetric matrix. Fix an orthonormal basis $\{v_k\}_{k=1,\dots,n}$ of eigenvectors. Then

$$A = \lambda_1 v_1 v_1^T + \cdots + \lambda_n v_n v_n^T.$$

Let A and $\{v_1, \dots, v_n\}$ be as described in the spectral theorem. What is the matrix A when written using this basis?